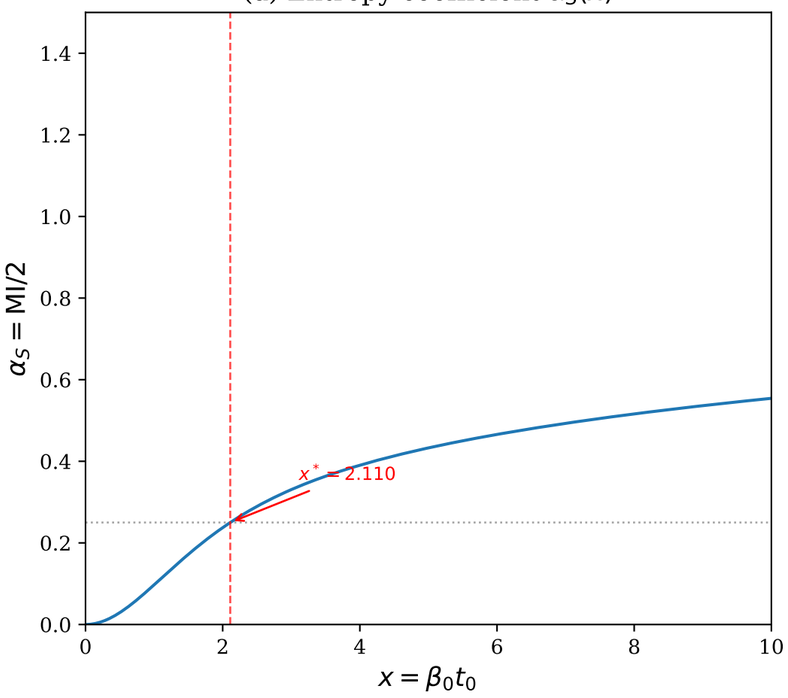
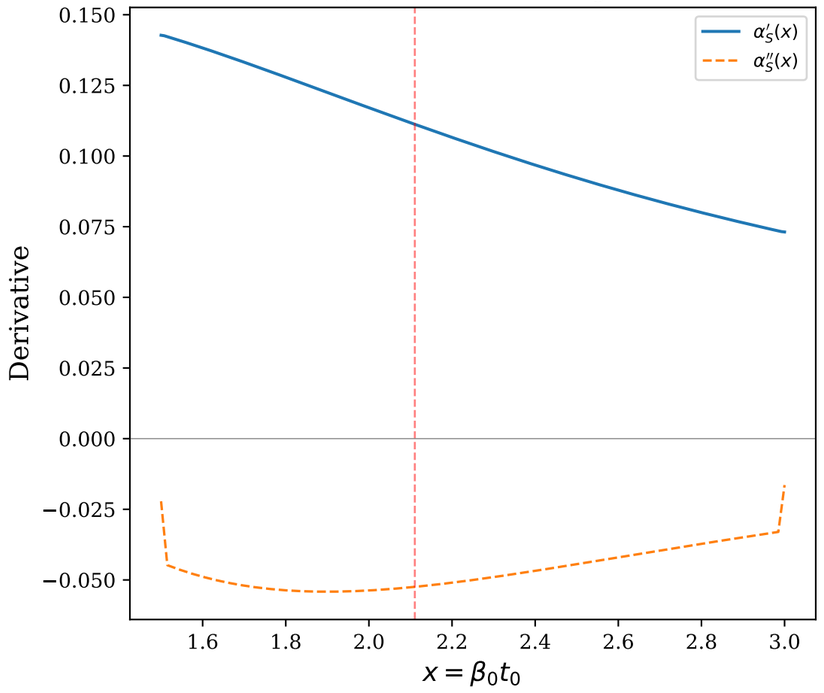
Thermodynamic stability analysis at $x^* = 2.1105$ (MI/2 = 1/4)

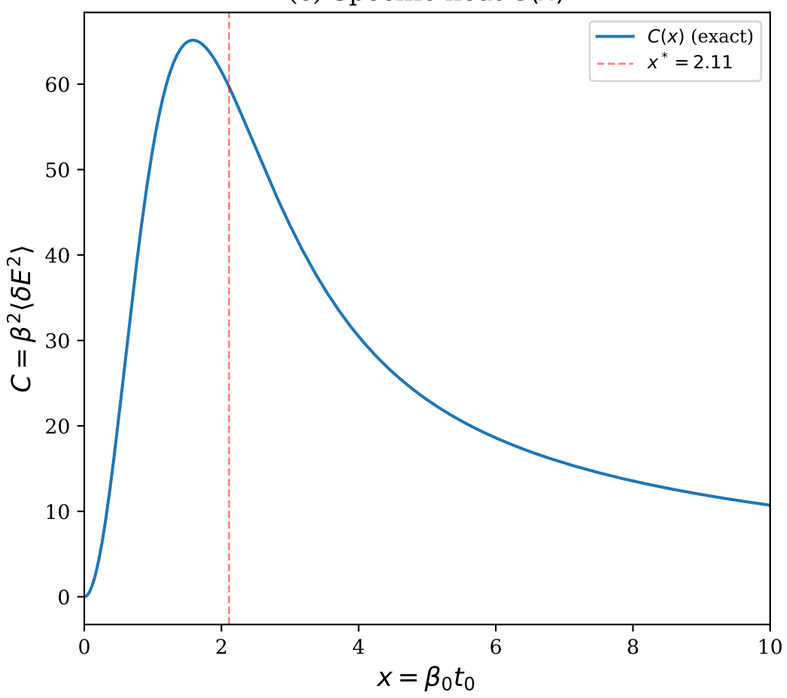
(a) Entropy coefficient $\alpha_S(x)$



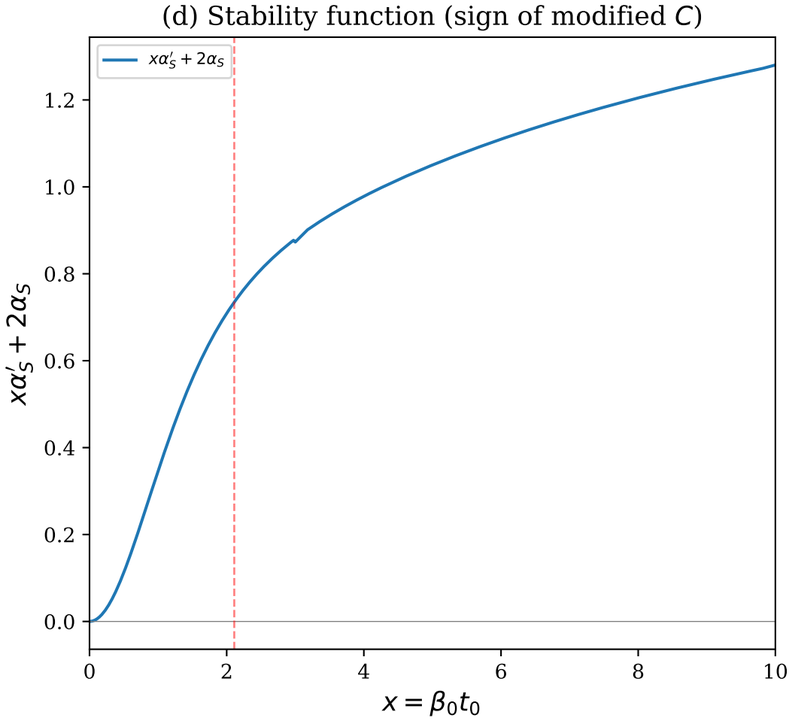
(b) Derivatives of α_S



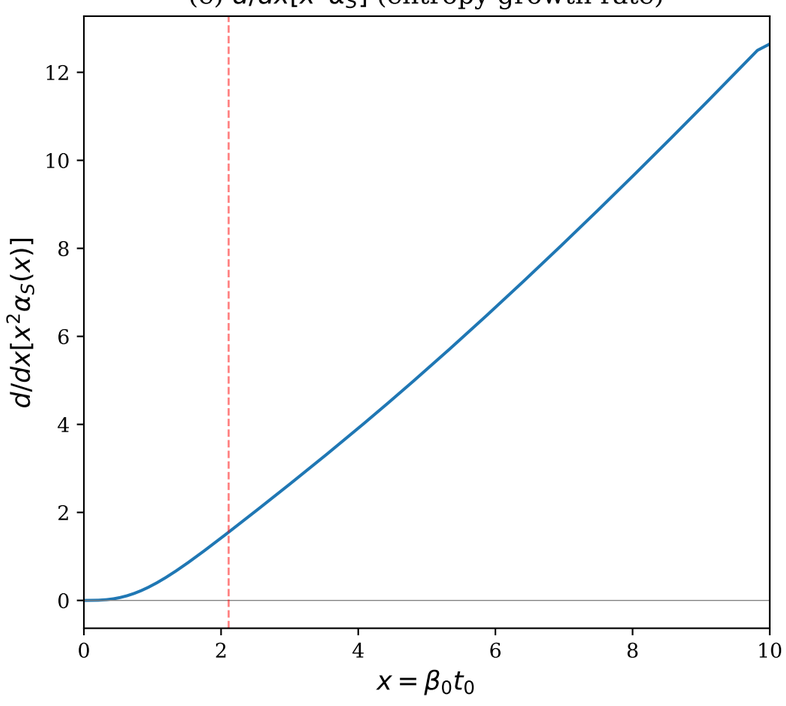
(c) Specific heat $C(x)$



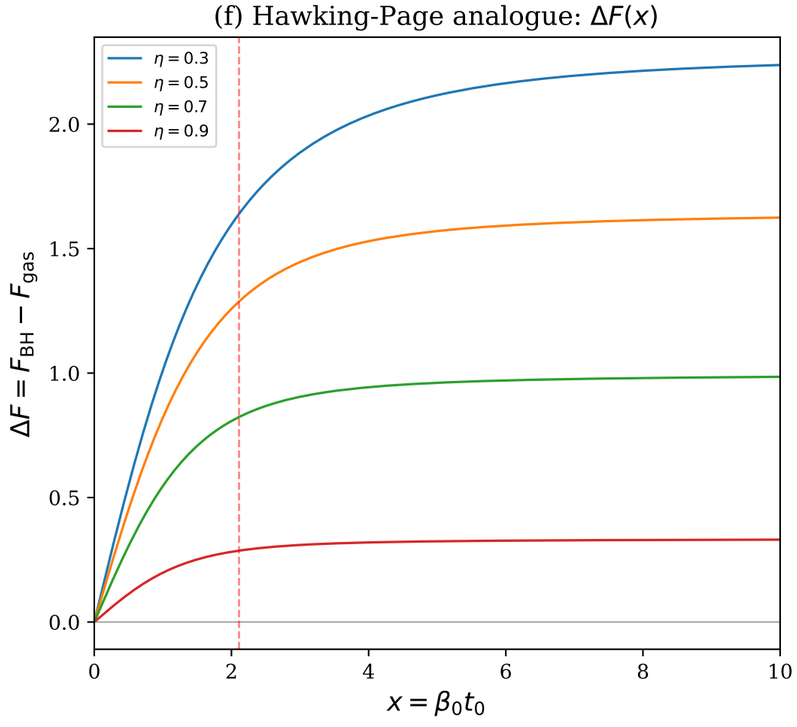
(d) Stability function (sign of modified C)



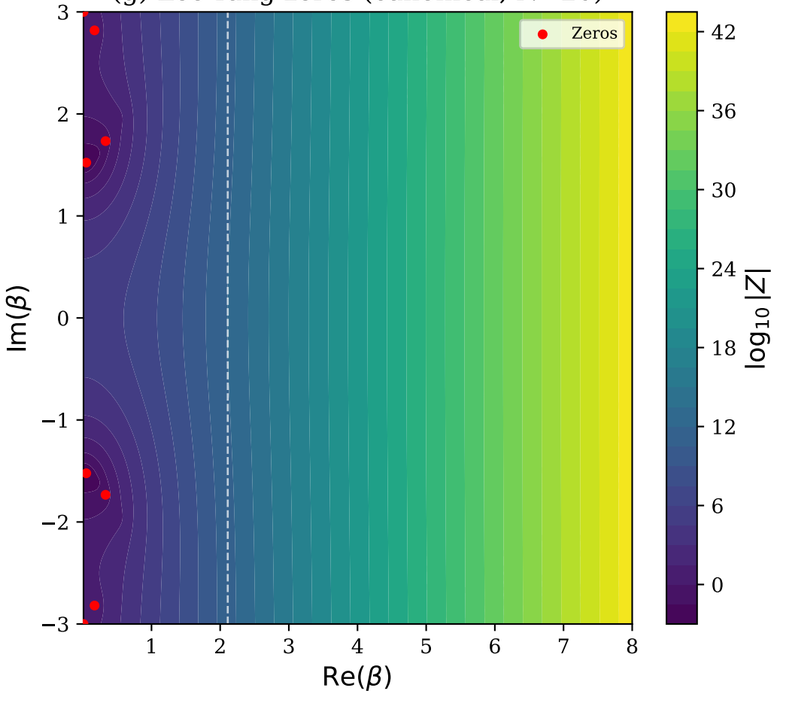
(e) $d/dx[x^2\alpha_S]$ (entropy growth rate)



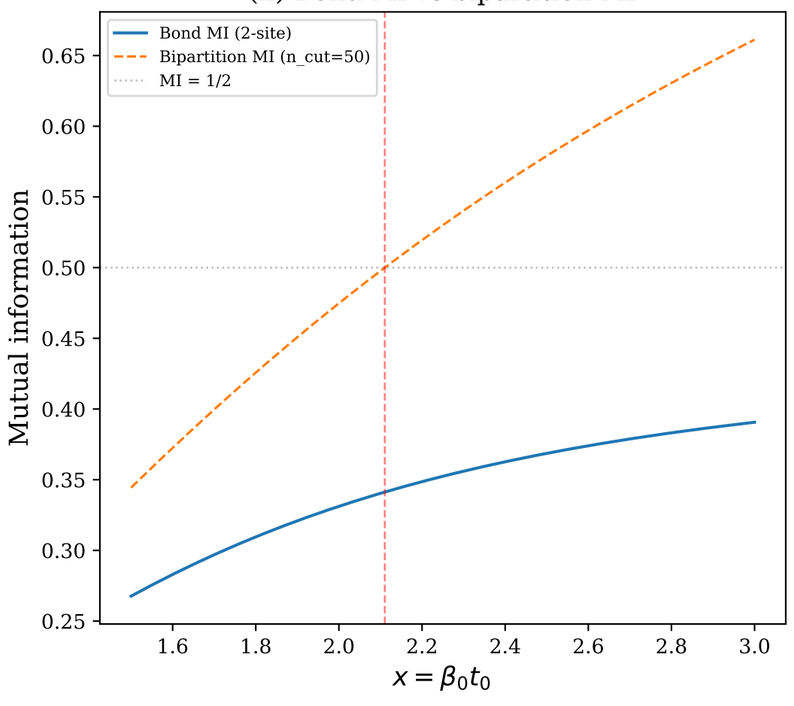
(f) Hawking-Page analogue: $\Delta F(x)$



(g) Lee-Yang zeros (canonical, N=20)



(h) Bond MI vs bipartition MI



(i) Correlation length and active modes

